CODE ALPHA TASK 1 ANALYSIS REPORT | QUERY PROTEIN: HEMOGLOBIN BETA (HBB) | SUBMITTED BY: Somya Mittal

A. Introduction:

Bioinformatics is an interdisciplinary field that integrates biology, computer science, and computational tools to analyze and interpret biological data. With the rapid increase in biological sequence data, bioinformatics has become essential for understanding the structure, function, and evolution of genes and proteins. Proteins are fundamental biomolecules that perform a wide range of biological functions, including catalysis, transport, and structural support. Understanding protein sequences helps in predicting their function and role in various biological processes. In this study, an in silico analysis was carried out on the HBB protein. Hemoglobin beta is a crucial component of hemoglobin, the oxygen-carrying protein found in red blood cells. It plays a vital role in transporting oxygen from the lungs to tissues and returning carbon dioxide back to the lungs. Mutations in the HBB gene are associated with several genetic disorders, including sickle cell anemia and beta-thalassemia, making it an important subject for bioinformatics analysis. The aim of this project was to analyze the protein sequence, validate its quality, perform similarity searches, and predict its functional role using computational tools.

B. Objectives:

=> The main objectives of this study were:

1. To retrieve the protein sequence from a biological database
2. To perform basic sequence analysis using computational tools
3. To validate the quality and integrity of the protein sequence
4. To identify homologous sequences using BLAST
5. To perform functional annotation using biological databases
6. To interpret the biological significance of the protein

C. Similarity Search Using BLAST:

=> The validated protein sequence was subjected to BLAST (Basic Local Alignment Search Tool) analysis to identify similar sequences in the database.

=> BLAST compares the query sequence with sequences in the database and provides results based on similarity. The key parameters considered include:

1. E-value (expectation value)
2. Sequence identity
3. Alignment score

=> The top hits obtained from BLAST included:

1. NP_000509.1
2. XP_018891709
3. AAX37051.1
4. AAX29557.1
5. AAN8454.1

=> These hits showed high similarity with hemoglobin beta proteins from different organisms.

D. Protein Sequence Retrieval:

The protein sequence of Hemoglobin Beta was retrieved from the NCBI in FASTA format. This database provides access to a vast collection of biological sequences and related information.

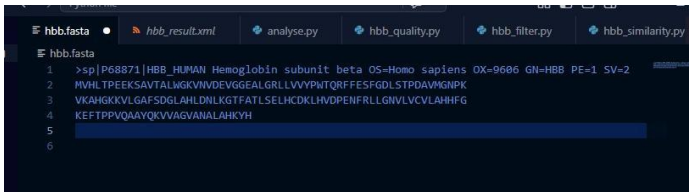
The retrieved sequence served as the input for all subsequent analyses.

E. Steps Involved in Analysis:

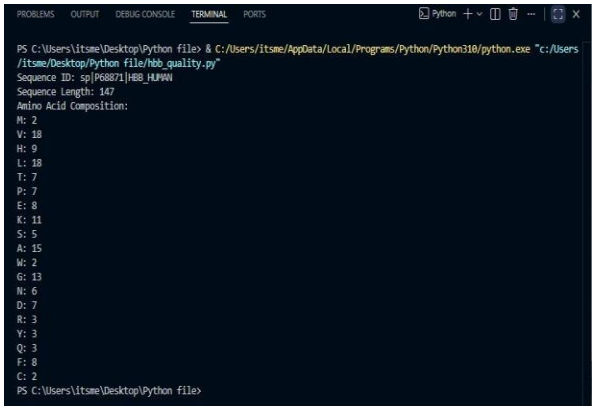
1. Biological Sequence Selection
2. Sequence Quality And Basic Analysis
3. Sequence Filtering and Validation
4. Homology Search (BLAST)
5. Functional Annotation
6. Biological interpretation

F. Results and Analysis:

1. Sequence that retrieved from UniProt Database in FASTA Format:

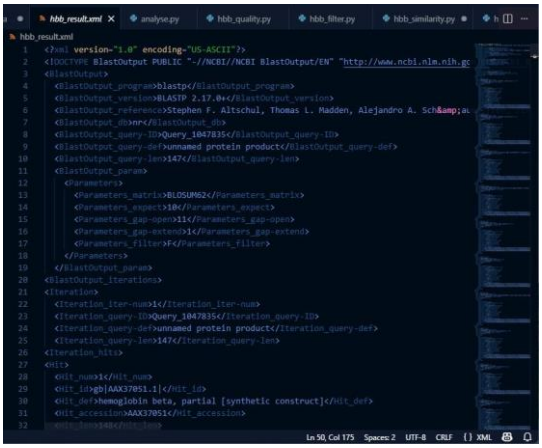


2. Quality Check Output: Sequence ID, Amino Acid Composition, Sequence length

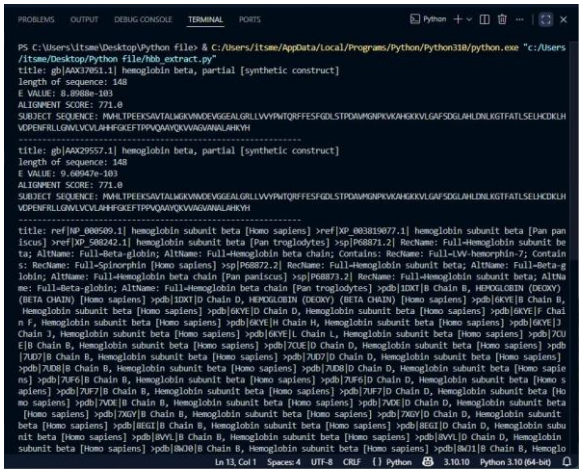


3. Sequence Validation And Filter Output: Checking if the Sequence is too Short for analysis or sequence is valid for not for the downstream analysis.

4. Similarity Sequences found in .XML format using BLAST:



5.Top 5 Best Similar sequences found by Reading .XML File output:



F.1 Result Interpretation:

=> The results obtained from sequence analysis and validation indicate that the selected protein sequence is of high quality and reliability. The absence of unknown residues and the appropriate sequence length confirm that the data is accurate and complete.

=> The amino acid composition further supports the functional nature of the protein, as key residues involved in oxygen binding and structural stability are present. These findings validate the use of this sequence for further analysis such as BLAST and functional annotation.